

AI ENGINEER

TECHNICAL ASSESSMENT

VectorDB • Embeddings • Knowledge Graphs • RAG

Take-Home Assessment • 48 Hours

Amunra Advisors LLP

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Assessment Overview

This assessment evaluates your ability to work with financial data in the context of building AI-powered research and analytics systems. It has two parts: Part A tests core concepts using clean data, and Part B tests real-world engineering ability using realistic data.

	Part A: Foundations	Part B: Real-World Engineering
Focus	Embeddings, vectors, basic knowledge graph	Data cleaning, entity resolution, chunking, full knowledge graph from mixed sources
Data	15 clean trade records (CSV)	20 trade records (CSV) + 3 analyst research notes (text)
Difficulty	Intermediate	Advanced
Weight	40% of total score	60% of total score

Deliverables

- **Jupyter Notebook(s)** with runnable code and markdown explanations between sections
- **Brief write-up** (in notebook or separate markdown) explaining your design choices
- **requirements.txt** or equivalent for environment reproduction
- Submit as a **zip file** or **GitHub repository link**

Guidelines

- You may use any Python libraries, embedding models, vector databases, and graph libraries of your choice.
- If using a paid API, document the model and explain why you chose it.
- Code quality matters — clean, readable code with comments.
- If you make assumptions, state them explicitly.
- **Do not use ChatGPT/Claude to generate the entire solution.** We will ask you to explain your code line by line in the follow-up interview.

Part A: Foundations

You are provided with **part_a_trades.csv** — a clean dataset of 15 trade records from 3 investment funds trading Indian equities and options on NSE.

Data Schema

Column	Description	Example
trade_id	Unique trade identifier	T001
date	Trade date (YYYY-MM-DD)	2025-01-15
fund	Fund name	Alpha Growth Fund
instrument	Ticker / instrument name	RELIANCE
exchange	Trading exchange	NSE
trade_type	BUY or SELL	BUY
quantity	Number of units	500
price	Trade price (INR)	2450.75
instrument_type	EQ (equity) or OPT (option)	EQ
sector	Industry sector	Energy
notes	Analyst notes on trade rationale	Initiated position post Q3 results...

Tasks

Task A1: Generate Embeddings

Choose an embedding model, generate embeddings for the trade data, and store them with metadata.

Task A2: Store in a Vector Database & Query

Load embeddings into a vector database and run the following queries. Show top-3 results with similarity scores for each:

	Query
Query 1	"Which trades were related to banking sector weakness?"
Query 2	"Show me all protective hedging activity"
Query 3	"What IT sector positions were taken and why?"

Task A3: Build a Basic Knowledge Graph

Extract entities and relationships from the trade data to build a knowledge graph. Visualize it and write 2 example queries that the graph can answer but vector search alone cannot.

Part B: Real-World Engineering

You are provided with two data sources: **part_b_trades_messy.csv** (20 trade records) and **part_b_research_notes.txt** (3 analyst research notes covering banking, IT, and Reliance Industries).

Tasks

Task B1: Data Cleaning & Entity Resolution

Clean the trade data and resolve entity inconsistencies across the dataset. Output a clean DataFrame and document your cleaning decisions.

Task B2: Chunk & Embed the Research Notes

Design a chunking strategy for the research notes, generate embeddings, and store them alongside the trade embeddings in a vector database. Run the following query:

Query

"What is our banking sector view and which funds have banking exposure?"

Task B3: Build a Comprehensive Knowledge Graph

Build a knowledge graph that integrates both the structured trade data and unstructured research notes. Visualize it and demonstrate 3 graph traversal queries that answer investment questions. At least one must be a multi-hop query.

Task B4: Design Reflection (Written)

In 300-500 words, address the following:

- If this system processed 10,000 research documents and 1 million trades, what would you change? What would break first?
- How would you evaluate whether your retrieval system is returning good results? What metrics would you track?
- When would you use the knowledge graph vs. vector search vs. a direct SQL query? Give one example for each.

Questions about the assessment? Email your point of contact at Amunra Advisors. Good luck!